DELAY-EMBEDDING ANALYSIS OF NEURAL NETWORK TRAINING LOGS: PRECONDITIONS, NEGATIVE RESULTS, AND VALIDATED DRIVER RECOVERY

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ABSTRACT

Empirical dynamic modelling promises to recover the structure of a high-dimensional system from scalar observations, which makes it attractive for monitoring neural network training from logs that consume negligible memory. This report establishes when that promise is redeemable. The embedding theorems of Takens and Stark reconstruct a compact invariant set that the trajectory revisits. We measure recurrence directly and find it absent over a whole run, then test the weaker hypothesis that short windows are locally stationary, logging every optimisation step to make that testable. The training loss does recur locally, but the binding constraint is the number of independent observations, which no logging rate changes: the correlation time is 163 steps against a transition of 1615, so the entire transition affords ten independent samples whether rows are written every step or every tenth. We then report three candidate signals for the delayed generalisation phenomenon known as grokking, each of which separates training runs convincingly and none of which survives a controlled comparison: an intrinsic-dimension estimate that reproduces closed-form constants of a straight line, a function-space velocity that tracks weight decay rather than generalisation, and a shared-manifold coupling that tracks configuration rather than the transition. Against these we set a positive result obtained in the regime where Stark's theorem does apply. Convergent cross mapping recovers an externally injected driver from a single one-dimensional loss log in seven of eight coupled runs across two architectures, with no false positives among the control runs whose driver is logged but never applied, and with a limitation for strictly periodic drivers that we characterise quantitatively. Testing the reverse cross map, which the ground truth of these runs fixes by construction, shows that the dominant direction is recovered in seven of eight coupled runs while controls converge in neither direction, and also that unidirectionality is not demonstrable when the driver acts on the loss without delay. Finally, a factorial study separating the training split from the initialisation shows that the delay is itself a broadly distributed random variable, with thirty runs of one configuration spanning 1290 to 5600 steps and a single run of the same configuration reaching 11 790.

Keywords: empirical dynamic modelling, delay embedding, surrogate data, convergent cross mapping, grokking, training diagnostics.

1 INTRODUCTION

A training run exposes very little of itself cheaply. Weights and activations are large and expensive to store, whereas scalar logs such as the training loss, the validation loss and the parameter norm cost almost nothing. If the geometry of the underlying optimisation could be reconstructed from those scalars, monitoring would become inexpensive and architecture independent. This is the premise of empirical dynamic modelling, and it rests on the embedding theorems of Takens (Takens, 1981) and, for forced systems, of Stark (Stark, 1999; Stark et al., 2003).

The premise is attractive for grokking, the delayed generalisation reported by Power et al. (2022), where a scalar precursor of the transition would be valuable and where the weight norm is an obvious candidate observable. We evaluate that candidate and two others, establish where delay embedding is and is not licensed on training logs, and report a positive result in the regime where it is.

The organising claim is the following. The embedding theorems reconstruct a compact invariant set that the trajectory revisits. A training run driven by an external signal with its own attractor satisfies this requirement by construction. The intrinsic optimisation transient does not. Results that respect this boundary reproduce across architectures and controls; results that cross it recover artefacts of trajectory smoothness or of run configuration. Section 2 states the requirements every claim in this report is held to. Section 3 then measures the precondition, section 4 reports three signals that violate it, section 5 reports the positive result obtained inside it, and section 6 measures how variable the transition itself is.

2 Validation requirements

Delay-embedding statistics return a finite number on data satisfying none of the hypotheses behind them, and three of the estimators used below proved circular, mis-specified or underpowered in exactly that way. We fix the following requirements in advance and hold every claim here to them. Each costs well under an hour of computation.

1. **Measure the precondition before invoking a theorem.** If the trajectory does not revisit its past states, the language of attractors does not apply whatever an estimator returns. Section 3 finds the precondition unmet over a whole run, and section 3.1 finds it met locally for one observable of the three.

2. **Calibrate on a system with a known answer, against a reference matched in length and in sampling density.** A Lorenz attractor grows by 1.66 across the embedding range at $n = 12\,000$ and by 8.86 at $n = 2000$, so a reference of the wrong length manufactures a separation by itself (Fig. 3b). At $dt = 0.01$ the same attractor fails a determinism test at $p = 0.400$ and only at $dt = 0.1$ passes at $p = 0.010$. A negative result means nothing unless a matched reference is positive.

3. **Justify preprocessing per series.** Subtracting a moving average manufactures structure in a sparse series, and differencing removes the slow drivers one may be looking for: the periodic drivers of section 5 cross map at 0.09 and 0.12 differenced and at 0.81 and 0.80 raw.

4. **Test against surrogates, with an ensemble large enough to express the threshold.** A statistic that does not exceed an amplitude-adjusted Fourier ensemble reports the linear autocorrelation of the series. With 39 surrogates the smallest attainable rank p-value, $1/40 = 0.025$, coincides with the Bonferroni threshold for two nulls and leaves no room for the correction; we use 199. Neither null has power in the strictly periodic case (section 5.3).

5. **Include a control differing in one respect only.** The ghost drivers of section 5 are the model: identical architecture, schedule, loss trajectory and logging, differing only in whether the coupling exists. Controls differing in several respects at once cannot discriminate, which is how each signal of section 4 came to look convincing.

6. **For a causal claim, test the reverse direction.** A cross map in one direction establishes coupling, not causation. Run both at a matched embedding and read direction from convergence, not skill, which section 5.4 shows reaches 0.57 in the false direction.

7. **Hold configuration fixed, vary only seeds, and record the expected sign in advance.** A predictor of the transition must correlate with the delay across runs of one configuration. This test falsified the most promising candidate in section 4.

3 Preconditions for delay embedding

Takens' theorem and its extensions guarantee that a delay map is an embedding of an *invariant set*. The guarantee is asymptotic and geometric: the orbit must return near its past states, so that neighbours in the reconstructed space are dynamically related rather than merely adjacent in time. When this fails, nearest-neighbour statistics measure the local tangent to the trajectory instead of the geometry of an attractor, a distortion first identified by Theiler (1986).

We therefore measure recurrence before invoking any theorem. For a delay-embedded series we fix a neighbourhood radius once, as a low quantile of all pairwise distances, and report the recurrence rate as a function of a temporal exclusion window. Fixing the radius once is essential; recomputing it after each exclusion would force the recurrence rate to equal the quantile for any input whatsoever, an error we made and corrected during this work. The shape of the resulting profile separates the two cases of interest. On an invariant set the rate falls to a positive plateau, because genuine returns occur at all temporal separations beyond the orbit's period. On a transient the rate decays towards zero, because every close pair was adjacent in time.

The estimator is calibrated on systems with known answers. A Lorenz-63 trajectory retains 0.93 of its recurrence rate at the widest exclusion, whereas a monotone ramp, which has no dynamics at all, decays to exactly 0.00. Figure 1 applies it to the training logs.

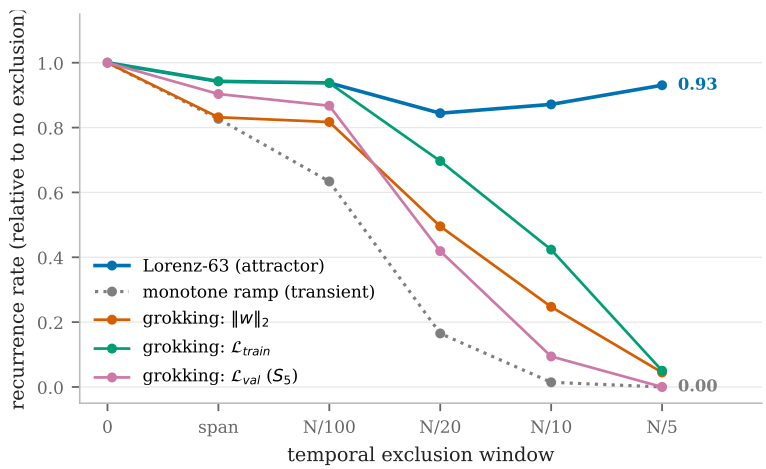

Figure 1: Recurrence rate as a function of the temporal exclusion window, normalised by the rate at zero exclusion. The Lorenz attractor holds a plateau, while all three training observables decay towards the monotone-ramp reference. Over a whole run the trajectories do not revisit their past states, so no single invariant set is available for the embedding theorems to reconstruct. Section 3.1 examines whether shorter windows escape this, and finds that the training loss does recur locally while the weight norm does not.

All three training observables behave like the transient reference rather than like the attractor. The validation loss of the S_5 runs reaches a ratio of 0.000 and the weight norm falls between 0.00 and 0.07.

That measurement is about the run as a whole, and by itself it settles less than it appears to. Failing to return to its past states over twenty thousand steps does not prevent a trajectory from being usefully stationary over a few hundred, and delay embedding applied inside such a window would be legitimate. The global measurement rules out treating a run as one invariant set; it does not rule out a local analysis. We therefore test the local hypothesis directly.

3.1 The sampling interval, and what it can and cannot fix

A local analysis has two requirements that pull against each other: the window must be short enough that the dynamics do not drift across it, and long enough to contain independent observations. At one row per ten optimisation steps these cannot both be met. A 300-sample window, the length at which the dimension estimate of section 4.1 is computed, spans 3000 optimisation steps against a median memorisation-to-generalisation gap of 2875: the window is longer than the transition it is meant to resolve.

We therefore re-ran three conditions logging every optimisation step. Only the inexpensive series need that rate, since the training loss is already computed by the update and the parameter norm is a reduction over the weights, whereas a validation pass is a forward pass over the held-out split; the two are logged on separate schedules, and a test asserts that neither the logging rate nor a skipped evaluation perturbs training.

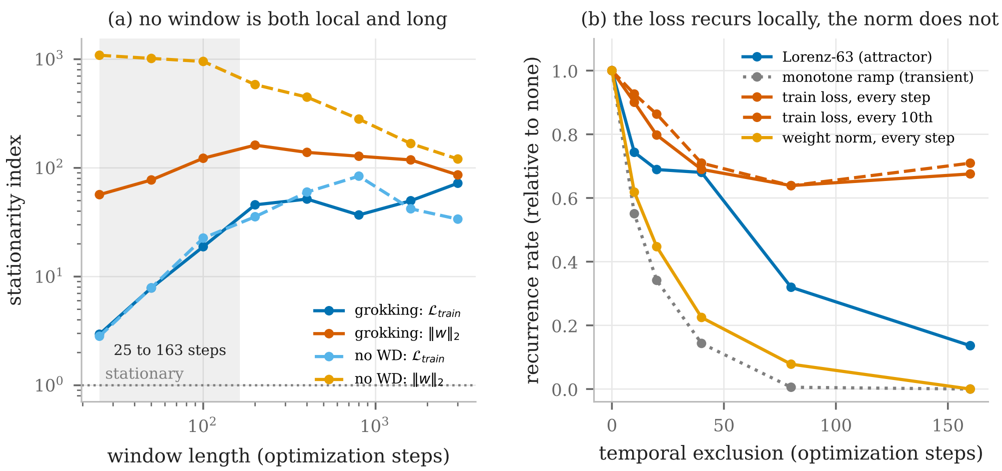

Figure 2: (a) Stationarity index against window length, in units of the value white noise of the same length produces, so unity means indistinguishable from stationary. The normalisation is necessary: the unnormalised ratio falls as $1/\sqrt{L}$ for a stationary series, so a fixed threshold would report long windows as more stationary by arithmetic alone. The shaded band runs from the longest stationary window, 25 steps, to one correlation time of the training loss, 163 steps. (b) Recurrence inside an 800-step segment against temporal exclusion. The training loss plateaus near the Lorenz reference while the weight norm decays to the ramp, and logging ten times more densely leaves both unchanged.

Three results follow. First, and correcting the impression left by Fig. 1, *the training loss does recur locally*: inside an 800-step segment its profile plateaus between 0.64 and 0.69 against a Lorenz reference of 0.68, while the monotone ramp decays to 0.00. The claim that training logs never revisit past states holds globally and fails locally. Second, *the weight norm does neither*, decaying like the ramp and never falling below a stationarity index of 20 at any window length, against 1 for a stationary series. Third, *density was not what was missing*: the dense and decimated profiles in Fig. 2(b) agree throughout.

What binds is the number of independent observations, which is the window length divided by the correlation time and so depends on neither the logging rate nor our choices. Inside the transition the training loss has a correlation time of 163 steps against a gap of 1615, giving ten independent samples whether a row is written every step (1615 rows) or every tenth (161 rows). Against this, Fig. 3(b) puts the length needed to resolve even a Lorenz attractor at $n \gtrsim 6000$. The two constraints close from both sides: the longest window indistinguishable from stationary is 25 steps, shorter than a single correlation time, so no scale is both stationary and observed more than once independently.

The autocorrelation agrees. The dense training loss falls to 0.76 at lag one and is then flat near 0.72, an independent minibatch-sampling term of about a quarter of the variance on a slowly varying component, rather than dynamics hidden below the previous resolution; a Lorenz trajectory at comparable density instead decays smoothly through 0.87, 0.61, 0.38, 0.23. Consistent with this, no training observable exceeds an IAAFT surrogate ensemble under simplex projection, the grokking training loss scoring 0.739 against a surrogate mean of 0.722 at $p = 0.52$. That comparison is only interpretable against a reference at comparable density, as section 2 requires: Lorenz at $dt = 0.01$ is oversampled in the same way and also fails, at $p = 0.400$, while at $dt = 0.1$ it passes at $p = 0.010$.

One caveat constrains recurrence as a diagnostic. Independent noise also produces close pairs at arbitrary separations and scores near unity, so recurrence separates smooth transients from everything else but not an attractor from noise, which is why the determinism test above accompanies it.

4 SIGNALS THAT DO NOT SURVIVE CONTROLLED COMPARISON

4.1 INTRINSIC DIMENSION OF THE WEIGHT-NORM TRAJECTORY

The Levina–Bickel maximum likelihood estimator (Levina & Bickel, 2004) applied to a sliding window of the weight-norm series produces a curve that rises during memorisation and falls before generalisation. Four observations remove its interpretation as a dimension. The first is analytic.

Proposition 1 (The estimate is a constant of a straight line)**.** *Let the delay vectors be sampled at uniform intervals from a curve that is locally straight at the scale of the neighbourhood, and let a Theiler exclusion of W samples be applied. The k nearest neighbours of an interior delay vector are then its temporal neighbours at offsets $\pm(W{+}1), \pm(W{+}2), \ldots$, so the ordered neighbour distances satisfy $r_j \propto s_j$, where s is the non-decreasing sequence $W{+}1, W{+}1, W{+}2, W{+}2, \ldots$ truncated to k terms. The Levina–Bickel estimate*

$$\hat{d} \;=\; \left[\tfrac{1}{k-1}\sum_{j=1}^{k-1}\log(r_k/r_j)\right]^{-1} \;=\; (k-1)\Big/\sum_{j=1}^{k-1}\log(s_k/s_j) \tag{1}$$

is therefore a function of k and W alone, and contains no information about the data. For $k = 5$ it equals 1.330 when $W = 0$ and 10.765 when $W = 14$.

Figure 3(a) verifies this. On a synthetic straight line the estimator returns the predicted constant to within 1.1% at every k and W tested, and on the two control runs trained without weight decay, which never generalise, it returns the same constants: medians of 1.33 and 1.38 without exclusion against a prediction of 1.33, and 11.30 and 11.69 with exclusion against 10.76. Both plateaus are reproduced by a straight line.

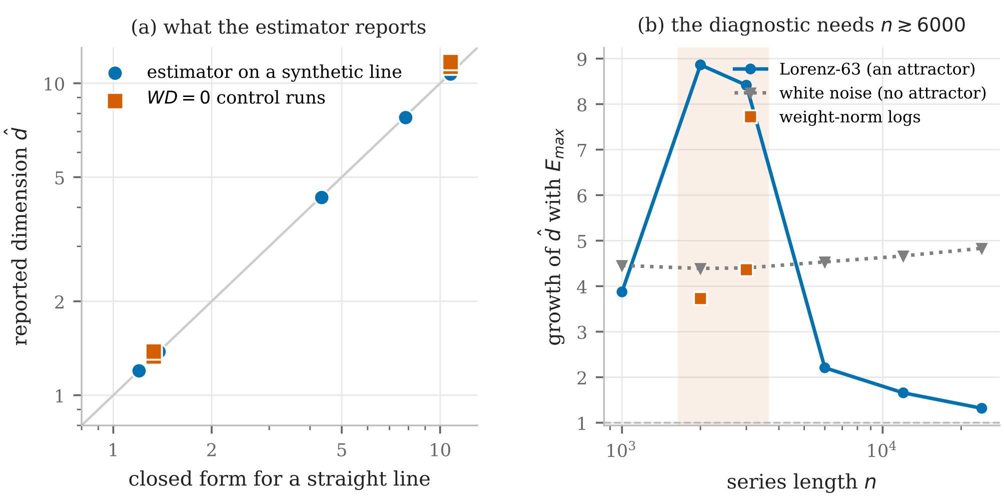

Figure 3: (a) Dimension reported by the estimator against the closed form of Proposition 1, on logarithmic axes with the identity line shown. (b) Growth of the estimate between $E_{max} = 5$ and $E_{max} = 30$, against series length. A value near unity is the signature of a resolvable attractor; the Lorenz reference reaches it only beyond $n \approx 6000$, and inside the shaded band, the length training logs provide, it scores worse than both white noise and the logs.

The second observation is that no dimension is identifiable at this sample size. An intrinsic dimension exists only if the estimate is insensitive to the space it is measured in, but that diagnostic is itself strongly length-dependent: a Lorenz reference grows by 8.86 at $n = 2000$, 2.21 at $n = 6000$ and 1.32 at $n = 24\,000$, while white noise holds between 4.4 and 4.8 (Fig. 3b). At the lengths available it therefore has no power, since a textbook attractor scores worse than the weight-norm series it should outperform, at 8.86 against 3.73 and 4.36. We draw no conclusion about whether these logs resemble noise; what follows is a bound on the data budget, and section 2 states it as such.

The third observation is a controlled comparison, and it does not depend on a Theiler exclusion, which the estimate is often computed without. Proposition 1 at $W = 0$ returns 1.330, so switching the exclusion off does not restore information; it pins the estimate to the tangent constant, which is why such curves look stable. That disposes of the reported *level* but not of the *shape*, since a control whose norm never leaves a straight line is flat by construction and cannot refute a claim about a rise and a fall. The control that can holds weight decay at 1.0 and reduces the training fraction, so the same regulariser drives a non-stationary norm in a model that never generalises. Figure 4 reports that comparison with no temporal exclusion at all.

The shape fails an internal check before reaching any control. Across three seeds of the identical configuration, whose weight norms end at 29.89, 29.12 and 30.39, the estimate falls in one run (Spearman trend -0.54) and rises in the other two ($+0.94$, $+0.90$), and their late-training levels are 1.51, 9.21 and 11.64, a spread of 7.7 within one configuration. The matched controls give $+0.75$ and $+0.22$, inside the range spanned by the runs that generalise.

What the estimate tracks is local smoothness. Pooled over 1026 windows it correlates at Spearman $+0.887$ with $\mathrm{std}(\Delta x)/\mathrm{std}(x)$, computed without any embedding, neighbour search or likelihood (Fig. 4c); within runs the correlation ranges from $+0.32$ to $+0.996$, so smoothness is a strong correlate rather than a complete description. The sign of the drift is immaterial, as expected of a smoothness statistic: the grokking norm falls from 42 to 30 and the lowdata norm rises from 42 to 61, and both collapse the estimate.

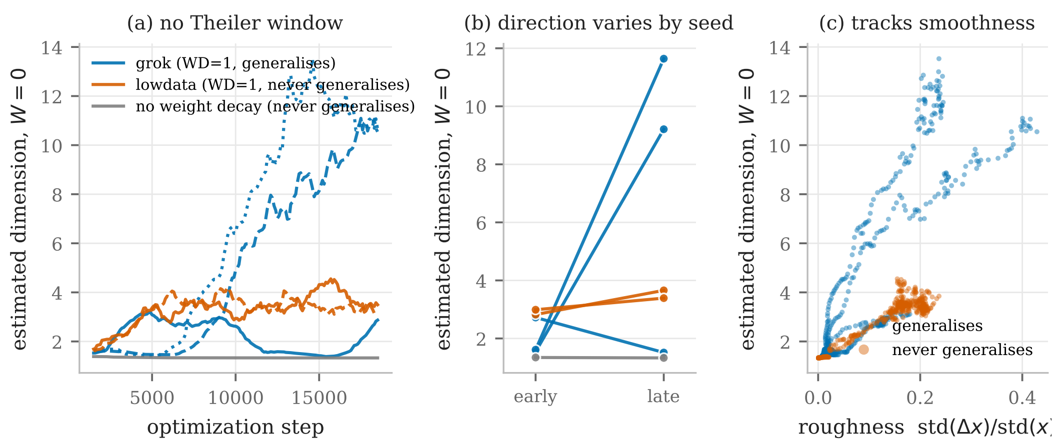

Figure 4: The dimension signal with no Theiler exclusion. (a) Sliding-window estimates; line style distinguishes seeds. (b) Median estimate over the first and last third of training. Two of three generalising runs rise, one falls, and the never-generalising runs lie between them; the no-weight-decay run is flat at the tangent constant, its apparent decline spanning only 1.34 to 1.33. (c) The estimate against a roughness ratio that involves no embedding.

The fourth observation, from section 3.1, is that the weight norm is never locally stationary at any window length from 25 to 3000 steps, with a stationarity index between 20 and 1087. No window of it can be read as a stationary system, so the estimate has no licensed interpretation even locally.

The third observation is procedural. The windows were labelled by their centres, so each estimate incorporated data recorded after the step it was assigned to. Under causal labelling, in which an estimate is assigned to the last step it observed, the reported lead over the transition does not survive.

Averaging the per-point estimates also inflates them relative to pooling the likelihood, as MacKay & Ghahramani (2005) note. The correction changes the level but not the conclusion, because the underlying quantity is not identifiable in the first place.

4.2 FUNCTION-SPACE VELOCITY

The preceding failure suggests an observable that cannot be a smoothness artefact of the weight norm. We therefore recorded, at each logged step, the logits of a fixed probe set of training inputs, centred each example's logit vector over classes and normalised it to unit length. The resulting series is invariant to the uniform rescaling of logits that weight decay induces, so a change in it indicates that the computed function has moved rather than that its confidence scale has changed. The median per-example movement of the normalised logits separates a grokking run from a run without weight decay by a factor of approximately 360.

The separation does not survive its control. We trained runs in which weight decay is held at 1.0 and only the quantity of training data is reduced, so that the model memorises and never generalises. Figure 5 shows the outcome. The runs that never generalise sustain a *higher* velocity than the runs that grok. The comparison carries three seeds of the grokking condition against four never-generalising runs, at two training fractions with the smaller replicated across three seeds, and the two families do not overlap. The only run far below the others is the one without weight decay, in which the function freezes within a few thousand steps and thereafter only rescales.

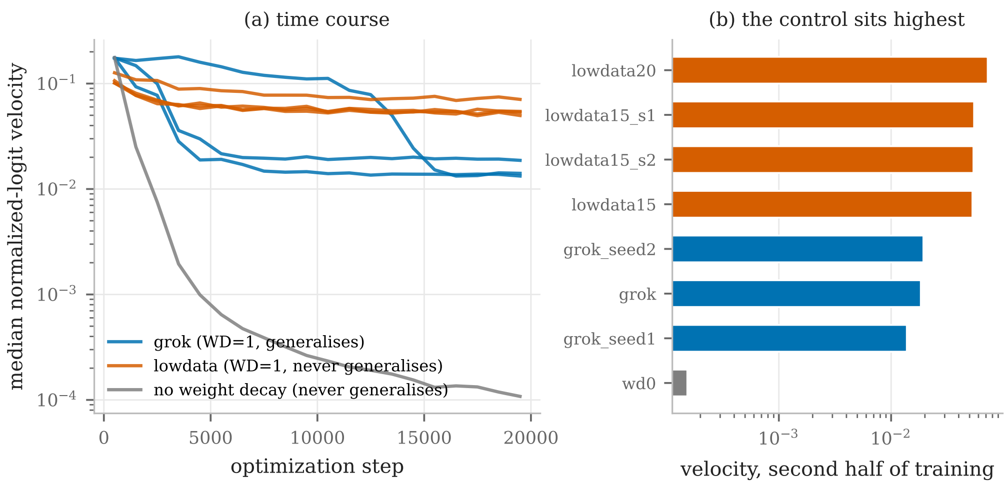

Figure 5: (a) Median velocity of the normalised probe logits against the optimization step, on a logarithmic scale. (b) The same statistic over the second half of training, one bar per run. If the velocity indicated impending generalisation, the runs that never generalise would sit low. They sit highest. Weight decay is identical in the blue and orange families, which differ only in the quantity of training data.

Quantitatively, the runs that never generalise span 5.30×10^{-2} to 7.26×10^{-2} over the second half of training against 1.38×10^{-2} to 1.94×10^{-2} for the grokking runs, a separation of a factor of 2.73 between the closest members of the two families. The three seeds of the reduced-data condition agree to within 3.4%, so the ordering does not rest on a single draw. The run without weight decay falls to 1.53×10^{-4}. Every run with weight decay lies within a factor of five of every other, and the factor of 360 quoted above is entirely accounted for by the presence or absence of regularisation. The statistic therefore measures the reorganisation that weight decay imposes on the function, and among regularised runs it is, if anything, inversely related to generalisation.

4.3 SHARED-MANIFOLD COUPLING

Both preceding failures are statistics of a single transient series. Cross mapping asks a different question, namely whether two observables lie on a common manifold, and that question is well posed on a transient because it requires no recurrence of either series alone. We therefore tested whether the training loss and the parameter norm, both available without any validation data, share a manifold, and whether the coupling changes at the transition.

The result appeared decisive. Figure 6(a) shows a clean separation with no overlap: the three runs with a delayed transition have median coupling between 1.00 and 1.15, while the four runs without lie between 2.87 and 5.61.

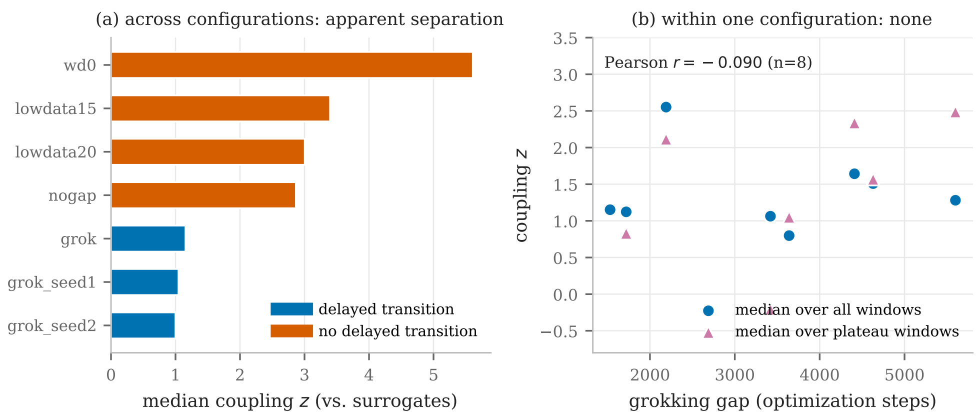

Figure 6: (a) Median coupling between the training loss and the parameter norm separates runs with a delayed transition from runs without. (b) The same statistic across eight runs of a single configuration, in which only the split and initialisation seeds vary. The gap spans 1290 to 5600 steps and the coupling is unrelated to it. The separation in panel (a) reflects configuration, not the transition.

Two checks dissolve it. Within individual runs the statistic does not move at the moment of generalisation; the apparent step in an aggregate over runs arises from averaging runs whose transitions occur at different steps. More importantly, the grouping in panel (a) coincides exactly with configuration, since the low group consists of the runs with a training fraction of 0.3 and weight decay of 1.0 and the high group contains every other setting. We therefore held configuration fixed and varied only the split and initialisation seeds, which the study of section 6 shows produce gaps between 1290 and 5600 steps. The expected sign was recorded before the runs were executed. The observed correlation between the gap and the coupling is -0.090 (Pearson) and $+0.214$ (Spearman) over eight runs, that is, nothing, and for plateau windows it is weakly positive, which is the opposite of the prediction. Within one configuration the coupling spans 0.80 to 2.55, against 1.0 to 5.6 across configurations.

4.4 A COMMON FAILURE MODE

The three fail in the same way, as table 1 summarises. Each separates *configurations* rather than *transitions*, and each rests on a comparison in which the runs differ in more than the property of interest. We regard this, rather than any individual negative result, as the principal methodological lesson.

Table 1: The three signals, the comparison that made each convincing, and the control that falsifies it. In every case the falsifying control differs from the positive runs in one respect only.

Signal	**Apparent evidence**	**Falsifying control**	**What it measures**
Intrinsic dimension of $\|w\|_2$	Rises then collapses before generalisation	A straight line reproduces both plateaus; across seeds of one configuration the estimate rises twice and falls once	Local smoothness of the series
Function-space velocity	Separates $WD = 1$ from $WD = 0$ by a factor of 360	$WD = 1$ with reduced data never generalises, yet sustains a higher velocity	Reorganisation imposed by weight decay
Shared-manifold coupling	Separates runs with a delayed transition from runs without, with no overlap	Across eight runs of one configuration the coupling is unrelated to the delay, $r = -0.090$	Run configuration

5 DRIVER RECOVERY IN THE REGIME WHERE THE THEOREMS APPLY

Stark's extension of Takens' theorem covers a system driven by an external signal that possesses its own attractor (Stark, 1999; Stark et al., 2003). A training run modulated by an injected driver satisfies that hypothesis by construction, which makes it the natural setting in which to ask whether anything at all is recoverable from a one-dimensional log.

5.1 DESIGN

We use two independent settings. The first consists of existing logs from a ResNet-18 trained on CIFAR-10, in which a fraction of each batch's labels is corrupted according to a driver logged alongside the loss. The second reproduces the design in this work, using a one-layer transformer on modular addition with drivers under our control. In both cases the ground truth is known, and in both cases the essential control is a *ghost* driver, which is logged exactly like a real one while the batch is returned unmodified. A ghost run shares the architecture, the schedule, the loss trajectory and the logging cadence of a coupled run and differs only in whether coupling exists.

Detection uses convergent cross mapping (Sugihara et al., 2012): the response is delay embedded and used to predict the driver, on the reasoning that if the driver forces the response then the reconstructed manifold of the response carries information about the driver's state. Significance is assessed against two nulls. The first uses iterative amplitude-adjusted Fourier surrogates (Theiler et al., 1992; Schreiber & Schmitz, 1996), which preserve the power spectrum and the exact value distribution of the driver while destroying its alignment with the response. The second rotates the driver in time, which preserves every value and changes only the alignment. No result is reported without them.

5.2 RESULTS

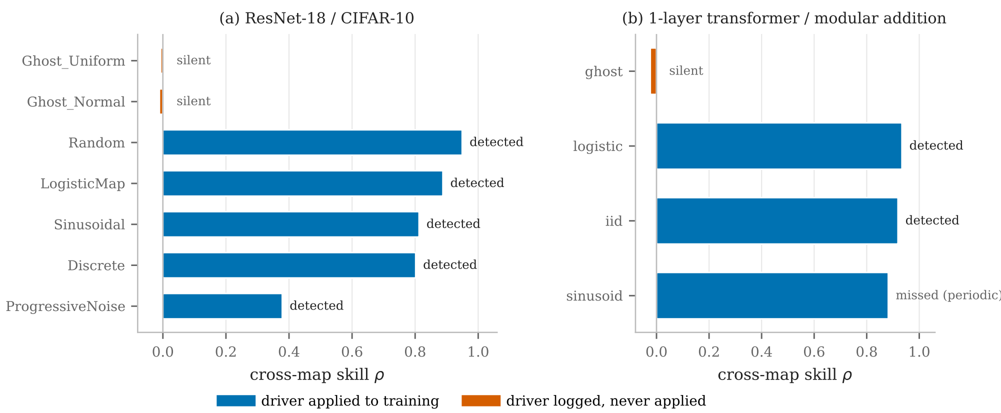

Figure 7: Cross-map skill for every run in both settings. Drivers that were applied to training are recovered from the loss log alone; drivers that were logged but never applied are not. The single missed detection is the smooth periodic driver discussed in section 5.3.

Figure 7 summarises both settings. All five coupled drivers in the ResNet setting are detected, with skills between 0.379 and 0.949 and $p \leq 0.05$ under both nulls; in the transformer setting the chaotic and independent drivers are detected at 0.934 and 0.919. Seven of eight coupled runs are detected and no ghost produces a false positive. Three of the five ghosts carry a varying driver and cross map at -0.011, -0.007 and -0.024; the other two carry a constant driver with no variation to recover, so they are controls but not evidence, and are omitted from Fig. 7.

The ghosts carry the argument. They share the training trend with the coupled runs, so their near-zero skill excludes the explanation that defeated every signal of section 4, that a statistic separates runs because their loss curves have different shapes.

The rule declares a driver recovered when either null rejects, a family of two tests whose false-positive rate approaches 0.10 rather than 0.05, and the 39-surrogate ensemble cannot express the correction because its p-value floor of 0.025 is the Bonferroni threshold itself. Repeating both experiments with 199 surrogates leaves every verdict unchanged: seven true positives, one false negative, no false positives under either rule. The one marginal case, the ResNet periodic driver at $p = 0.050$, resolves to $p = 0.015$.

Convergence with library size, the signature that gives the method its name and distinguishes causation from coincidental correlation, is shown for both directions in Fig. 8(a): skill rises for the coupled runs, most clearly for the independent driver at 0.59 to 0.95, while the ghosts show neither rise nor level. Differencing the response, the obvious way to remove the training trend, inverts this conclusion by removing the slow drivers with it (section 2); no manual removal is needed, since both nulls and the ghosts already control for the trend.

5.3 A LIMITATION FOR PERIODIC DRIVERS

The one missed detection is instructive and is not repaired by a better statistic. The smooth sinusoidal driver of the transformer setting yields a cross-map skill of 0.882, which is high, but its surrogates reach between 0.87 and 0.94, which is equally high. The reason is structural. The loss reveals the phase of the cycle, and any series of the same period, whether a spectrum-preserving surrogate or the driver rotated in time, is a deterministic function of that phase and is therefore equally predictable from it. Spectrum-preserving nulls consequently have no power against a strictly periodic driver. The corresponding driver in the ResNet setting is detected only because it is quantised to ninety-five distinct values, so its surrogates depart from it considerably further. A different null, for example a driver of a different period, would be required.

5.4 DIRECTION OF THE RECOVERED COUPLING

Detection is not direction: recovering the driver from the loss shows the two are coupled, not which forces which. Cross mapping is asymmetric and can be asked directly. Embedding Y and predicting X, written Y xmap X, is evidence that X drives Y (Sugihara et al., 2012), so the forward map is loss xmap driver. Here the ground truth is known rather than assumed, since the driver schedule is evaluated from a fixed seed before any gradient is taken and cannot depend on training.

The convention is calibrated on a system with a known answer, as inverting it would invert every causal statement resting on it. On coupled logistic maps with $\beta_{xy} = 0$ and $\beta_{yx} = 0.32$, so X drives Y alone, the true direction rises from 0.331 to 0.993 with library size while the false direction runs 0.585 to 0.569 and does not converge. The false direction reaches a skill of 0.57 while carrying no causal information, which establishes that convergence rather than skill is the discriminating quantity.

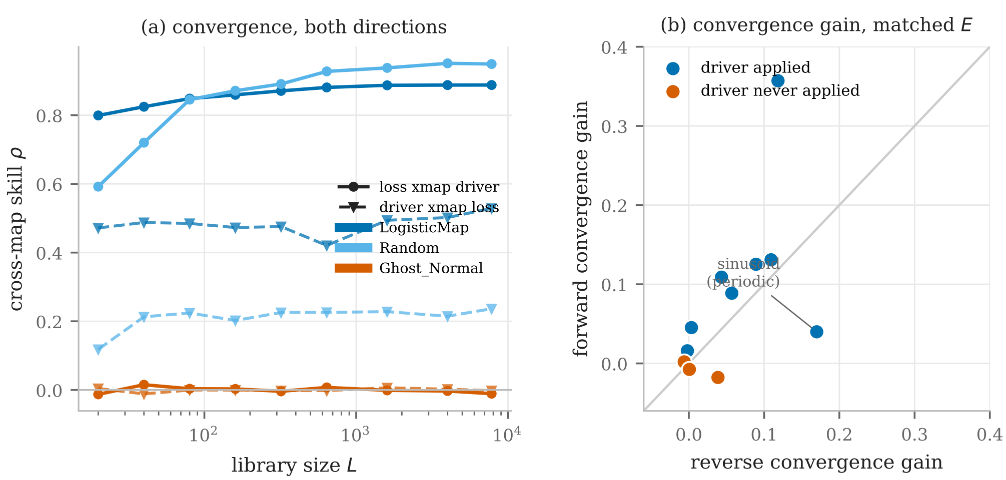

Figure 8: Forward against reverse cross mapping, both at a matched embedding dimension. (a) Skill against library size: the forward map converges for coupled runs while the reverse stays flat, and both are flat for the ghost. (b) Convergence gain. Coupled runs lie above the diagonal and ghosts sit at the origin; the exception is the periodic driver of section 5.3.

At a matched embedding the forward map converges more strongly than the reverse in seven of eight coupled runs, most clearly for the independent driver at gains of 0.357 against 0.118, while the ghosts converge in neither direction with all gains within 0.039 of zero. The exception is the periodic driver, whose reverse gain of 0.170 exceeds its forward 0.040; that run also fails forward detection at $p = 0.77$ and is excluded by section 5.3 on independent grounds.

A stricter reading does not support unidirectionality. Granting the reverse the best of five embedding dimensions rather than holding it at the forward's $E = 3$, it converges and beats its own null in most runs: no coupled run is then read as unidirectional and five of eight are read as bidirectional. Selecting a maximum over five embeddings inflates the reverse, but not by itself. The explanation is structural: the driver acts at every step, so the loss depends on it contemporaneously and a delay vector of the driver already contains the value the loss is partly a function of, letting the reverse map succeed by direct functional dependence with no manifold involved. Cross mapping does not separate causation from strong instantaneous coupling, the generalised-synchrony regime (Rulkov et al., 1995) known to compromise it (Ye et al., 2015).

We therefore claim the dominant direction, recovered correctly in seven of eight runs, and not unidirectionality. Establishing the latter would require a driver acting with a delay, so that the contemporaneous term is absent.

5.5 The method is not discriminative between a run's own metrics

For completeness we applied the same procedure to pairs of observables within single uninjected runs. Eleven of sixteen directed pairs pass both nulls, with skills as high as 1.000. This is consistent with the theory, since the training loss, the validation loss and the parameter norm are three observations of one system, but it means that cross mapping between a run's own metrics carries no information about the transition and should not be interpreted as a discovery.

6 Variability of the delayed transition

Any statistic proposed as a precursor of grokking is a statement about the delay, so the distribution of the delay is worth measuring before precursors are sought. The configuration seeds a single random stream from which the train and validation split is drawn and from which the model initialisation then continues. A change of seed therefore alters both which examples are trained on and what the initial weights are, and the two cannot be attributed separately. We separate them by restarting the stream between the two steps, verify that the modification is inert when unused and that it decouples the two factors in both directions, and run a factorial study of six splits by five initialisations with weight decay fixed throughout.

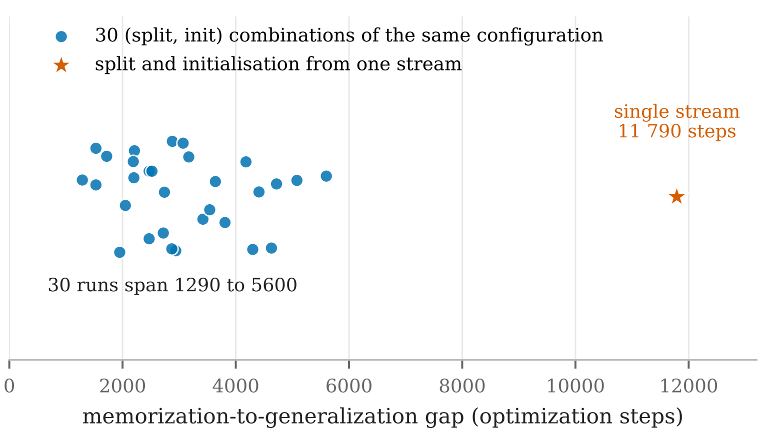

Figure 9: Memorisation-to-generalisation gap for thirty runs of one configuration, varying only the split and the initialisation seed. Every run generalises and every gap lies between 1290 and 5600 steps. Marked separately is a run of the same configuration in which the split and the initialisation are drawn from a single stream, whose gap of 11 790 steps lies far outside the distribution.

All thirty runs generalise, with gaps between 1290 and 5600 steps, a mean of 3062 and a standard deviation of 1130: within a configuration held otherwise fixed, the delay has a standard deviation exceeding a third of its mean.

A run of the same configuration under the coupled stream produced a gap of 11 790 steps, 7.7 standard deviations above the grid mean and 2.11 times its largest member, while two further runs at other seeds gave 1510 and 1570, both inside it. Five grid cells share the outlying run's split and give 2050 to 4300, so the split alone does not account for it, and no cell reproduces the pairing of split and initialisation the coupled stream produces; we attribute the outlier to that pairing. A delay measured from one run is therefore not a property of the configuration, and a precursor validated against one run has been validated against a single draw from a wide distribution.

The variance decomposition is inconclusive by design rather than by accident. The split accounts for 19.8% of the variance in the gap and the initialisation for 23.8%, with the remainder in the interaction. Neither factor dominates. A partial analysis of the first fifteen cells indicated the opposite, with the split accounting for 43% against 17% for the initialisation, which is a caution against reading factorial studies before they complete.

7 Conclusion

Delay-embedding methods apply to training logs in a narrower regime than their premise suggests, and within it they work. When an external driver with its own attractor forces training, convergent cross mapping recovers it from a single scalar loss log in seven of eight coupled runs across two architectures, with no false positives among the runs whose driver was logged but never applied, and the forward map converges more strongly than the reverse in seven of eight. These data do not establish unidirectionality (section 5.4). The quantity needs no validation data and comes from a log that costs nothing to store.

The intrinsic optimisation transient does not admit the same treatment, for a quantitative reason rather than a matter of taste: its correlation time is a tenth of the transition it would have to resolve, so the transition affords about ten independent observations at any logging rate, against the thousands an attractor reconstruction needs. The three statistics that appeared to detect the grokking transition separate run configurations instead, and the phenomenon is more variable than one run can reveal, thirty runs of a single configuration spanning 1290 to 5600 steps against 11 790 for a run of the same configuration seeded from one stream.

One question is left open: whether any training-side observable distinguishes a run that will generalise from one that will not, with configuration held fixed. None of the five examined here does, comprising the parameter norm, the trajectory roughness, the intrinsic dimension, the function-space velocity and the shared-manifold coupling. The driven regime is where the theory applies and validation against a known answer is possible, and it is where we would place further effort.

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